Graphic Era

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B TECH ECE V
Name of the Paper: ANALOG and DIGITAL COMM
Paper Code: TEC501
Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. What is modulation? Why is it necessary in communication systems? Compare and contrast amplitude modulation (AM) with frequency and phase modulation techniques. CO-1

b. Describe the generation and demodulation process of Double-Sideband Suppressed Carrier (DSB-SC) signals. Highlight its advantages and disadvantages compared to standard AM. CO-1

c. A 1 kHz sinusoidal signal modulates a carrier of frequency 100 kHz using amplitude modulation. CO-1
(i) Sketch the frequency spectrum of the modulated signal.
(ii) Calculate the bandwidth of the transmitted signal.

Q2. (2X10=20 Marks)

a. Derive the mathematical representation of Frequency Modulation (FM) and Phase Modulation (PM) signals. Discuss the relationship between FM and PM with an example. CO-2

b. Discuss the operation of a frequency discriminator used for demodulating FM signals. Explain its working principle with a neat diagram. CO-2

c. A Frequency-modulated signal is represented as: CO-2
$s(t) = 5 \cos [2\pi (10^5) t + 2 \cos (2\pi (10^3) t)]$
From the given expression:
(i). Find the carrier frequency and modulating frequency.
(ii). Calculate the frequency deviation and modulation index.

Q3. (2X10=20 Marks)

a. Describe the working of Pulse Amplitude Modulation (PAM). Discuss its advantages, limitations, and applications in modern communication systems. CO

b. A signal is given as: $x(t) = 10 \cos (3000\pi t) + 4 \sin (5000\pi t)$. Compute the Nyquist rate and Nyquist interval. CO-

c. Describe the working of Delta Modulation (DM). How does it differ from PCM and DPCM? CO-3

Q4. (2X10=20 Marks)

a. A baseband pulse transmission system has a channel with a bandwidth of 3 kHz. Determine the maximum data rate at which signals can be transmitted without ISI using the Nyquist criterion. Also, explain the significance of the result. CO-

b. Explain the working principle of Frequency Shift Keying (FSK). Highlight its advantages and disadvantages over other digital modulation schemes. CO-

c. Discuss the concept of Probability of Error in digital communication systems. Explain how it is evaluated for different digital modulation schemes like BPSK and QPSK. C

Q5. (2X10=20 Marks)

a. Discuss the concept of channel capacity and derive the channel capacity theorem for a discrete communication channel. CO-5

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b. Consider a binary symmetric channel with a crossover probability p = 0.1. Calculate the channel capacity C and interpret the result.

c. Explain the principle of error detection and correction in digital communication. Discuss the working of linear block codes with an example of encoding and decoding. CO